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Autoencoders in Multi-Omics Data Integration

Introduction to Bioinformatics
Section: A

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1 Introduction

Modern biology produces huge amounts of data from different sources, such as:

- Genomics (DNA)
- Transcriptomics (RNA)
- Proteomics (proteins)
- Metabolomics (metabolites)

They are called **multi-omics data** in together. Analyzing this type of data is difficult because:

- The datasets are extremely large
- Different omics types behave differently
- Some data is often missing
- Relationships between variables are very complex

Traditional statistical methods struggle with these challenges. Because of this, researchers now use deep learning methods, especially **autoencoders**, to analyze multi-omics data.

Autoencoders are powerful because they can:

- Reduce data size
- Learn hidden patterns
- Combine different omics datasets
- Handle noisy or incomplete data

This report explains how autoencoders and their advanced versions are used in multi-omics research.

2 What is an Autoencoder?

An **autoencoder** is a type of neural network that learns how to compress data and then reconstruct it.

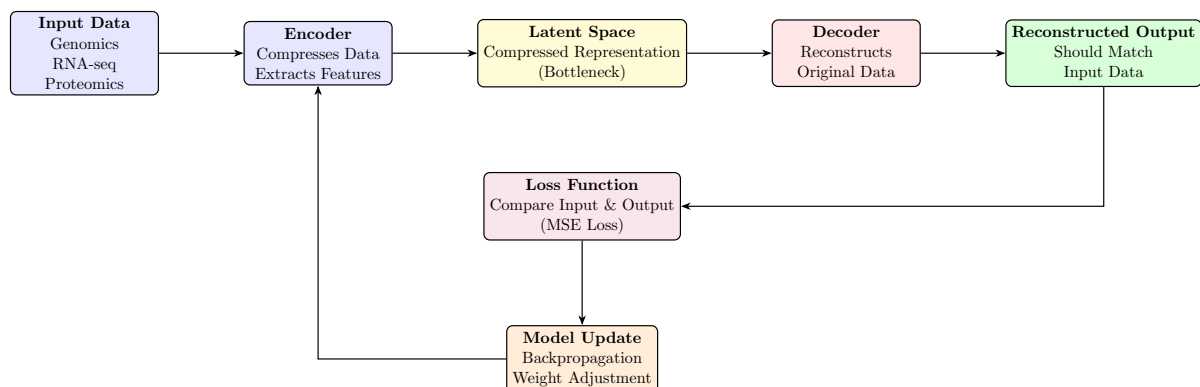


Figure 1: Basic Workflow of an Autoencoder

It has two main parts:

- Encoder: The encoder takes large complex data and converts it into a smaller simplified form called a **latent representation**.
- Decoder: The decoder tries to rebuild the original data from this compressed version.

The goal is to keep the most important information and to remove unnecessary Noises.

2.1 How Autoencoder Works

Working Process of an Autoencoder

- The encoder takes high-dimensional omics data as input.
- It learns important biological patterns from the data.
- The data passes through several neural network layers.
- Each layer reduces the size of the data step by step.
- Unimportant and noisy information is removed.
- Important features are compressed into a smaller representation.
- This compressed representation is called the **latent space** or **bottleneck**.
- The latent space stores the most useful biological information.
- The compressed data is then sent to the decoder for reconstruction.
- In multi-omics, the encoder helps combine genomics, transcriptomics, proteomics, and epigenomics into one unified representation.

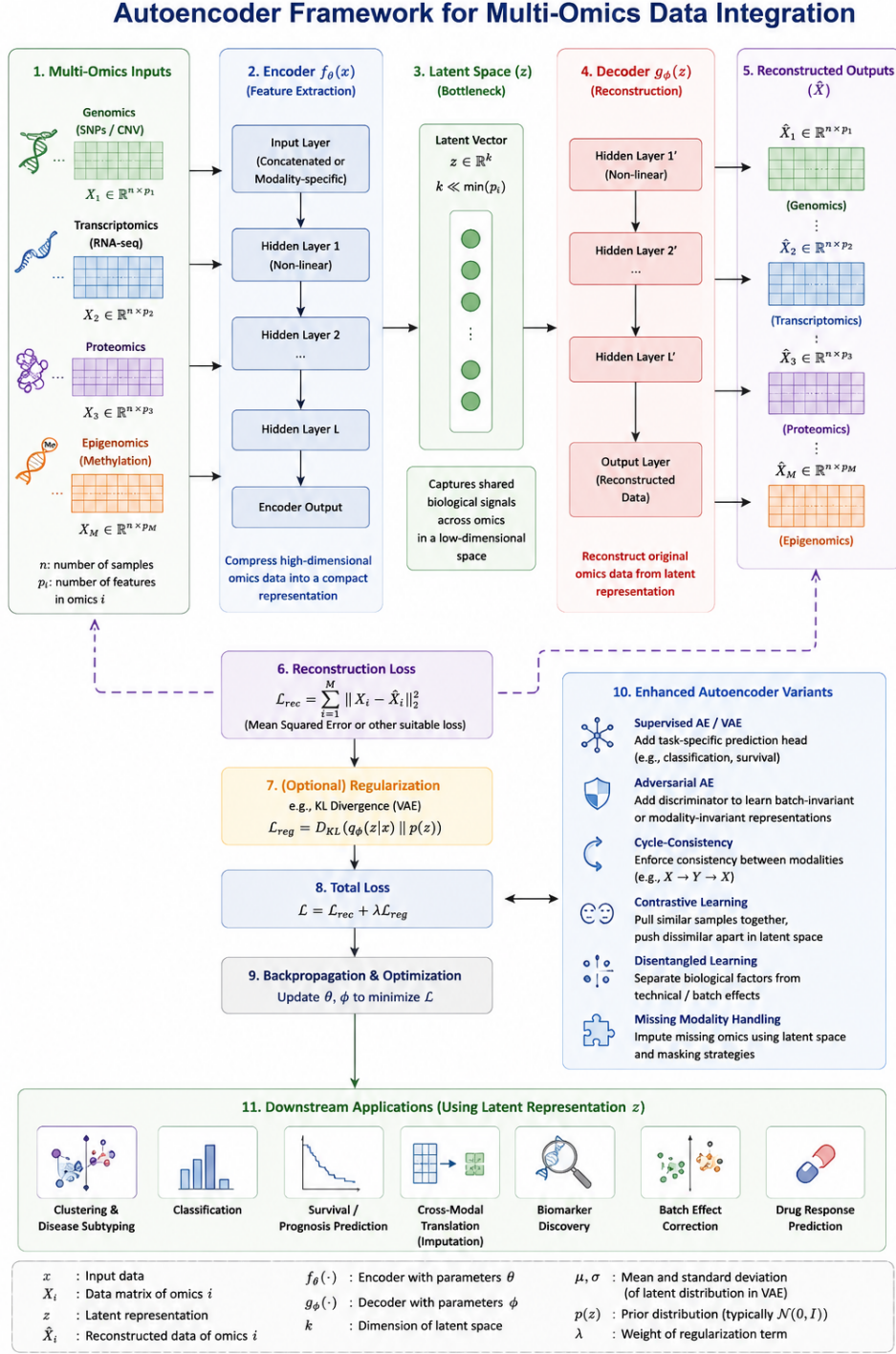


Figure 2: Autoencoder Architecture

2.2 Why Autoencoders are Useful in Multi-Omics

Multi-omics datasets contain thousands or millions of biological features. Autoencoders help by:

- Compressing high-dimensional data
- Finding important biological patterns

- Combining multiple omics types
- Supporting tasks like:
 - Disease prediction
 - Cancer subtype detection
 - Clustering
 - Missing data prediction

Unlike older methods such as PCA, autoencoders can learn non-linear relationships, which makes them much more powerful for biological data.

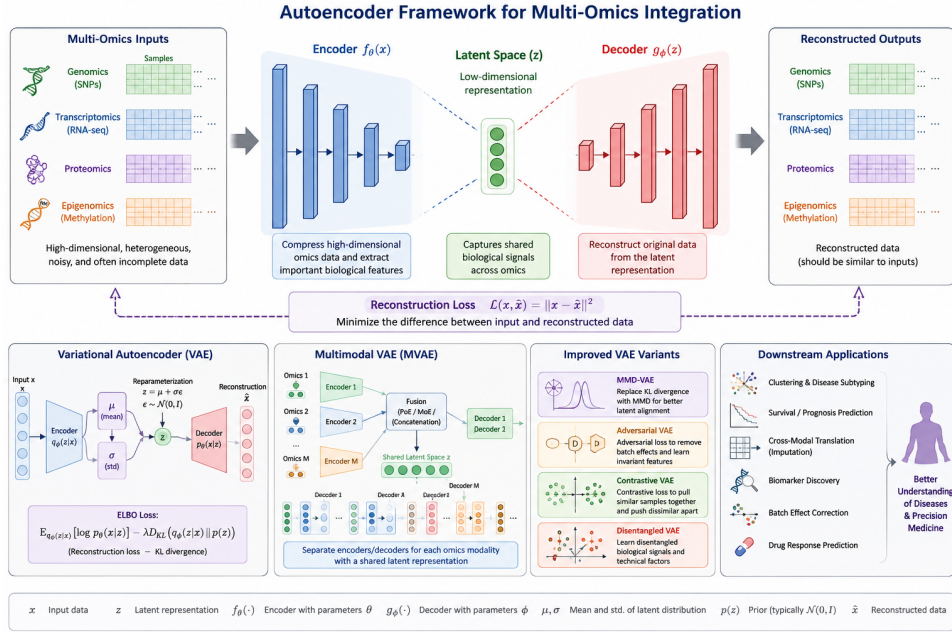


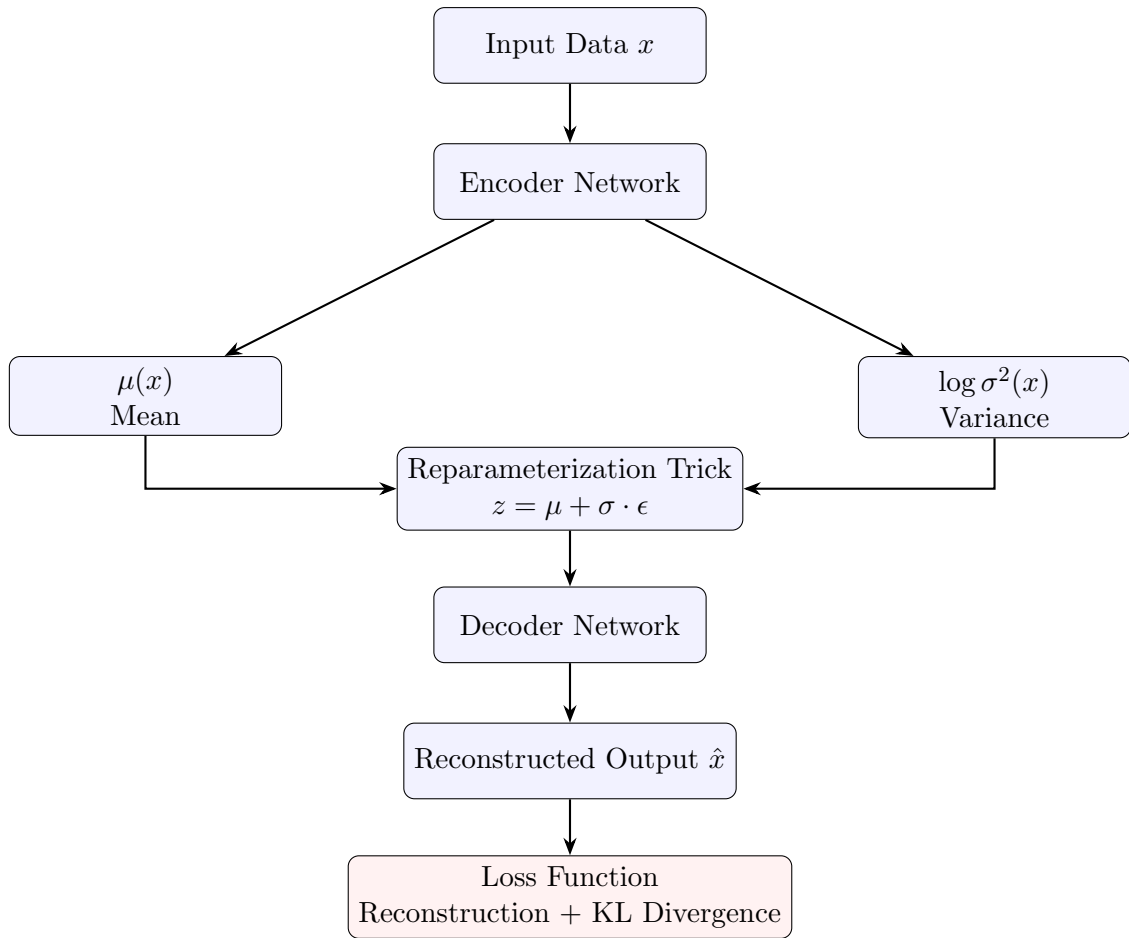
Figure 3: Autoencoder Architecture

3 Variational Autoencoders (VAEs)

A **Variational Autoencoder (VAE)** is a more advanced version of a normal autoencoder.

Instead of learning a fixed compressed value, VAEs learn a probability distribution. This means:

- The model understands uncertainty
- It can generate new synthetic data
- The latent space becomes smoother and more meaningful



3.1 How VAEs Work

In a VAE:

- The encoder creates:
 - Mean (μ)
 - Standard deviation (σ)

These describe a probability distribution.

The model then samples from this distribution to create the latent representation.

The decoder reconstructs the original data from this sampled latent vector.

Variational Autoencoder (VAE) – Flowchart

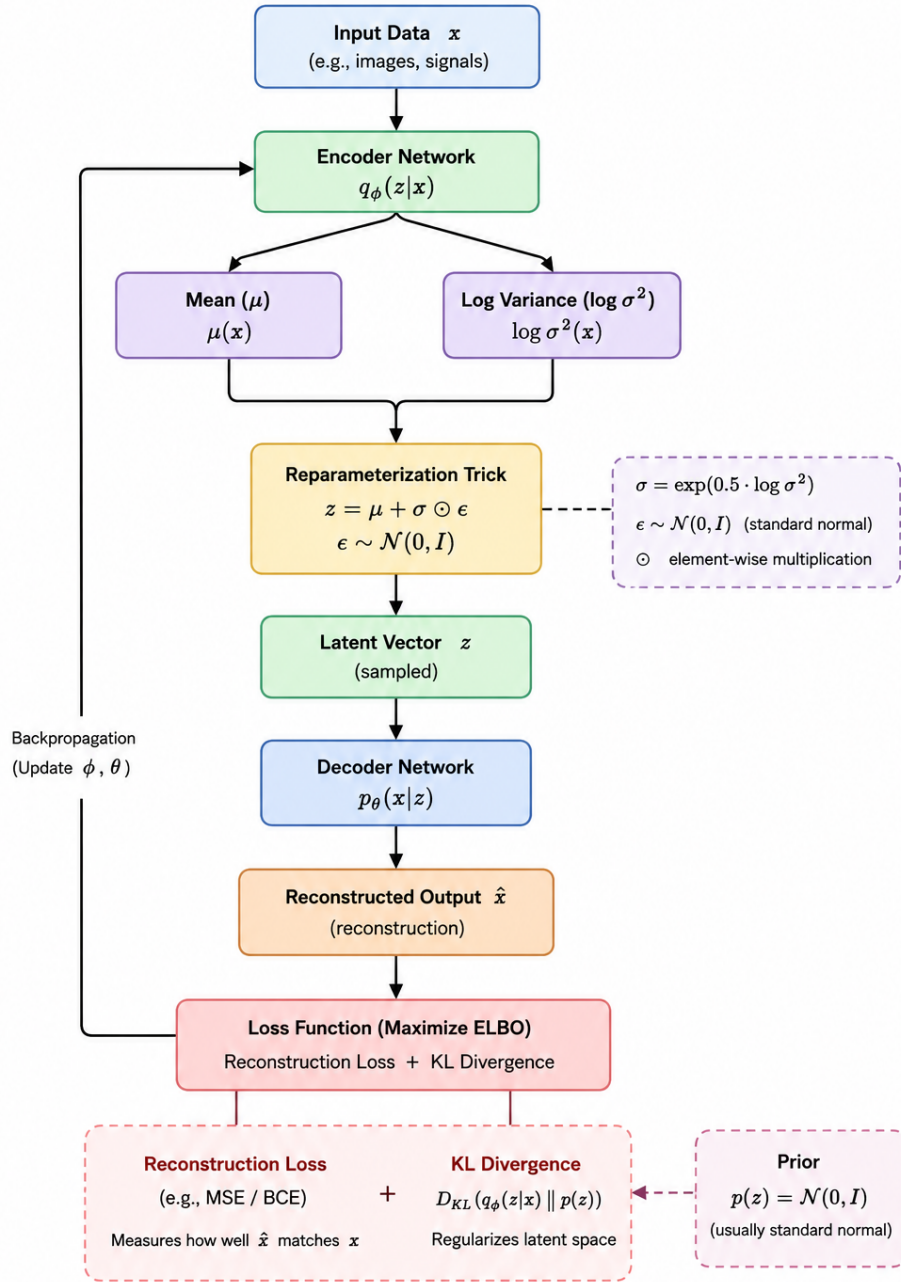


Figure 4: VAE Architecture

3.2 Why VAEs are Important

VAEs are very useful for multi-omics because they can:

- Handle noisy biological data: Biological datasets are often incomplete or inconsistent.
- Generate synthetic samples: Helpful when datasets are small.

- Learn meaningful latent spaces: This helps identify shared biological patterns across omics types.
- Support uncertainty estimation: Important in medical and biological applications.

4 Important Improvements in VAE-Based Models

To improve the performance of Variational Autoencoders (VAEs) in multi-omics data integration, researchers have proposed several methodological advancements. These improvements mainly focus on enhancing latent representation quality, handling multimodal data, improving interpretability, and addressing missing data issues.

4.1 Regularization and Latent Space Enhancement

A key challenge in VAEs is **posterior collapse**, where the model ignores the latent space. To address this, alternative regularization techniques have been introduced.

4.1.1 MMD Regularization

Maximum Mean Discrepancy (MMD) replaces or complements KL divergence to improve latent structure learning.

Benefits:

- Better latent representations
- More stable training
- Improved biological signal preservation

Representative Models:

- MMD-VAE
- Multigrade
- MultiVI
- scDisInFact

4.2 Supervised and Semi-Supervised VAEs

Standard VAEs are unsupervised, but biological datasets often include labels such as disease type or clinical outcomes. Incorporating supervision improves predictive performance.

Applications:

- Cancer classification
- Drug response prediction
- Survival analysis

Benefits:

- Higher classification accuracy
- Improved biological interpretability
- Better disease prediction performance

Representative Models:

- OmiVAE
- XOmiVAE
- OmiEmbed

4.3 Multimodal Integration Strategies

Multi-omics data comes from different biological sources. Multimodal VAEs are designed to integrate these heterogeneous datasets into a shared latent representation.

4.3.1 Fusion Strategies

- Concatenation-based fusion
- Product of Experts (PoE)
- Mixture of Experts (MoE)

Benefits:

- Captures shared biological signals
- Enables cross-omics integration

Representative Models:

- scMVAE
- scMM
- Multigrade
- MultiVI
- MIDAS

4.4 Cross-Modal and Coupled Representation Learning

Instead of forcing all modalities into a single latent space, some models maintain separate representations while learning relationships between them.

Benefits:

- Preserves modality-specific information
- Learns cross-omics relationships

Representative Models:

- DCCA
- JAMIE

4.5 Adversarial Learning Approaches

Adversarial training introduces a discriminator to align distributions across modalities or remove batch effects. The encoder learns to fool the discriminator, improving alignment.

Targets of Discrimination:

- Data modality
- Batch/source effects

Benefits:

- Better cross-modal alignment
- Reduced batch effects

Representative Models:

- SCIM
- GLUE
- scCross
- omicsGAN

4.6 Cycle-Consistency Learning

Cycle-consistency ensures that translating data between modalities and back preserves the original information.

Benefits:

- Improved cross-modal translation
- Higher reconstruction consistency
- More reliable embeddings

Representative Models:

- scMVP
- sciCAN
- Portal

4.7 Contrastive Representation Learning

Contrastive learning optimizes representations by pulling similar samples together and pushing dissimilar samples apart.

Example principle:

- Same tissue samples $\rightarrow$ closer in latent space
- Different tissue samples $\rightarrow$ farther apart

Benefits:

- More biologically meaningful embeddings
- Improved clustering and downstream prediction

Representative Models:

- MOSA

4.8 Disentangled Representation Learning

Biological data often contains both true biological signals and technical noise. Disentanglement methods separate these factors in the latent space.

Benefits:

- Cleaner biological interpretation
- Improved batch correction
- Increased model robustness

Representative Models:

- scDisInFact
- MIDAS

5 Handling Missing Data in Multi-Omics Integration

In real-world datasets, not all samples contain all omics modalities. This setting is known as **mosaic integration**.

5.1 Common Strategies

- Masking missing modalities during training
- Zero-padding incomplete inputs
- Using prior-based imputation techniques

5.2 Benefits

- Enables learning from incomplete datasets
- Improves robustness in real-world clinical data
- Allows prediction of missing omics modalities

Representative Models:

- Multigrade
- scVAEIT
- MIDAS

6 Popular Autoencoder Models

Model	Main Idea	Main Use
OmiVAE	Supervised variational autoencoder with label guidance	Cancer classification
MMD-VAE	Uses Maximum Mean Discrepancy instead of KL divergence for better latent structure	Cancer subtype prediction
OmiEmbed	Multi-task VAE for joint representation learning	Survival prediction
scMVAE	Multimodal VAE with shared latent space	Clustering and imputation
Multigrade	Product-of-Experts based multimodal integration with MMD regularization	Mosaic multi-omics integration
MultiVI	Joint embedding across multiple omics modalities	Batch effect correction
GLUE	Graph-based generative model for regulatory network learning	Gene regulatory inference
MOSA	Contrastive learning-based representation model	Biomarker discovery
MIDAS	Disentangled representation learning framework	Batch correction and imputation

Table 1: Popular Autoencoder-Based Models in Multi-Omics Integration

7 Challenges

Even though these models are powerful, they still have problems.

7.1 High Computational Cost

Deep learning models need:

- GPUs
- Large memory
- Long training times

7.2 Limited Data

Good multi-omics datasets are expensive and rare. Small datasets can cause:

- Overfitting
- Poor generalization

7.3 Interpretability Problems

Deep learning models are often difficult to explain biologically. Researchers still struggle to understand:

- What each latent dimension means
- Why the model makes certain predictions

7.4 Hyperparameter Tuning

Performance depends heavily on:

- Latent space size
- Loss weights
- Network design

Finding the best settings is difficult.

8 Future Directions

Researchers are now exploring:

8.1 Foundation Models

Large pretrained biological AI models such as:

- scGPT
- scFoundation

8.2 Integration Beyond Omics

Future systems may combine:

- Multi-omics data
- Medical images
- Electronic health records
- Wearable sensor data

This could greatly improve precision medicine.

8.3 Better Missing Data Handling

Future models will better support:

- Incomplete datasets
- Large patient populations
- Real-world clinical data

8.4 Better Interpretability

Researchers want models that are:

- More transparent
- Easier to explain
- More clinically trustworthy

9 Conclusion

Autoencoders and Variational Autoencoders (VAEs) are key deep learning models for multi-omics data integration, enabling efficient representation learning from high-dimensional and heterogeneous biological data.

They are widely used to:

- Compress and represent large-scale omics datasets
- Integrate multiple biological data modalities
- Support disease prediction and stratification
- Handle missing or incomplete data
- Capture complex, non-linear biological relationships

To improve their performance in real-world biological applications, several extensions have been developed, including MMD-based regularization, adversarial training, contrastive learning, disentanglement methods, and multimodal fusion strategies.

These advancements have significantly improved model robustness and biological relevance, making VAEs an essential component of modern computational biology. Their continued development is expected to further advance precision medicine, biomarker discovery, and personalized healthcare.